

Haoxuan Xie

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A short bio

I am a fourth-year PhD candidate at Nanyang Technological University, advised by [Siqiang Luo](#). My research focuses on **database systems** and **scalable data processing**. Practical examples related to my research include how to store data up to billion-scale, or how to handle data with dynamic workloads, etc. I also have some familiarity with LLM-related topics, such as vector database for Retrieval-Augmented Generation (RAG). You can refer to my publications to see my detailed research fields.

🎓 Education

Nanyang Technological University 2023.8 – Now

- PhD candidate in College of Computing and Data Science;
- Research fields: graph storage, graph algorithms and vector database;
- Supervisor: [Siqiang Luo](#);
- Expected graduation date: June 2027.

The Chinese University of Hong Kong, Shenzhen 2019.9 – 2023.7

- BEng (Computer Science and Engineering);
- cGPA (cumulative GPA): 3.704/4 (7th out of 102); mGPA (major GPA): 3.84/4 (6th out of 102);
- 2020, 2021 Dean's List; 2020, 2021 Class C Academic Performance Scholarship.

⚙️ Publications

Vector database:

- To appear in VLDB 2027 conference: [VectorMaton: Efficient Vector Search with Pattern Constraints via an Enhanced Suffix Automaton](#)

Graph storage:

- SIGMOD 2026 conference paper: [RadixGraph: A Fast, Space-Optimized Data Structure for Dynamic Graph Storage](#)

Graph algorithms:

- VLDBJ 2026 journal paper: [Efficiently Querying Connected Components in Large Temporal Graphs via Scalable and Maintainable Indices](#)
- ICDE 2025 conference paper: [Finding Near-Optimal Maximum Set of Disjoint k-Cliques in Real-World Social Networks](#)
- SIGMOD 2023 conference paper: [On Querying Connected Components in Large Temporal Graphs](#)

Key-value store:

- VLDB 2026 conference paper: [ArceKV: Towards Workload-driven LSM-compactions for Key-Value Store Under Dynamic Workloads](#)

♥️ Competitions and Awards

- Programming Contests: **Silver** medalist in ICPC 2021 East Continent Regional Contest (Jinan site); **Bronze** medalist in ICPC 2023 Asia Pacific Regional Contest (Vietnam site, ranked 16th out of 125 teams);
- Maths: the National First Prize in The Chinese Mathematics Competitions, 2020;
- Mathematical Modeling: the National Second Prize in Contemporary Undergraduate Mathematical Contest in Modeling, 2020; finalist award in The Interdisciplinary Contest in Modeling, 2021 (top 1%).

Experiences

Tencent IEG Internship

2022.12 – 2023.6

- Worked in the company, Tencent, as a researcher in the Interactive Entertainment Group;
- Developed algorithms for fair A/B testing, aiming to partition users into groups with similar interests while minimizing inter-group interference;
- Developed practical graph algorithms for a LOLM event, and published them as [a co-authored paper](#) in ICDE 2025 conference supervised by [Wenqing Lin](#).

CUHK(SZ) Research Assistant

2021.9 – 2022.10

- Performed research on graph algorithms supervised by the professor, [Yixiang Fang](#);
- Proposed a new problem: connected components in temporal graphs, which models real-world relationships and devise novel algorithms for the problem;
- Awarded with the Undergraduate Research Awards in CUHK(SZ);
- Published [a paper](#) as the first author in the SIGMOD 2023 conference.

PingCAP Internship

2022.5 – 2022.9

- Worked in the company, PingCAP, as a developer of distributed database and data migration tools;
- Contributed to the [TiDB](#) project and its corresponding data migration tool [TiFlow](#) project by identifying and fixing internal errors;
- Contributed to the backend of [TiDB Cloud](#) project by developing monitoring metrics on Grafana;
- Developed a mock DB named [DM simulator](#) to simulate the data migration process for testing new features.

HKU Research Internship

2022.7 – 2022.9

- Performed research on densest subgraph problems supervised by the professor, [Reynold Cheng](#);
- Formulated a new definition of practical densest subgraph problem and propose an approximation algorithm.